AI-Driven Smart Logistics & Supply Chain Intelligence System

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Course: AI for Logistics & Supply Chains

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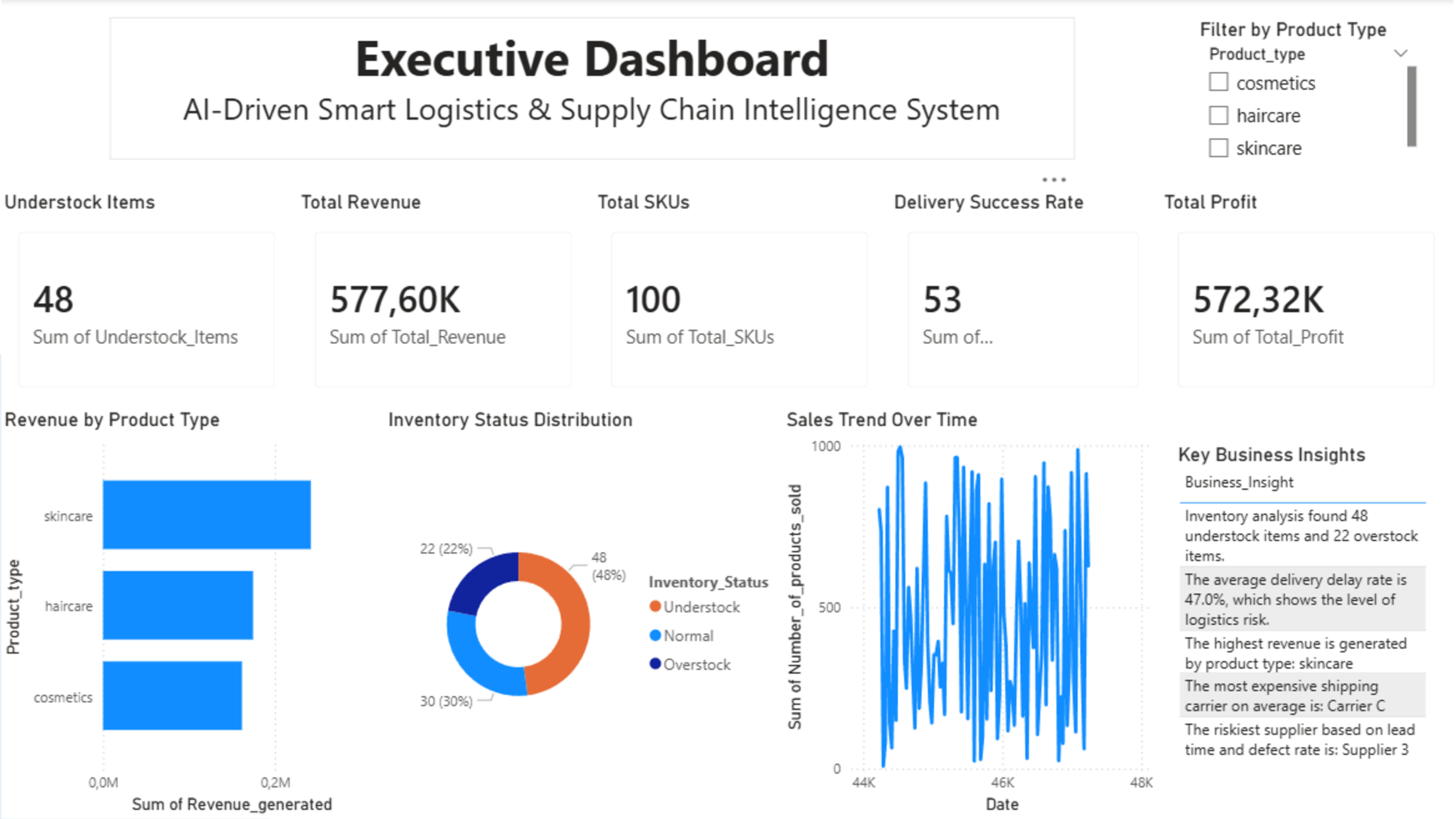
Project Overview

This project presents an AI-driven logistics and supply chain intelligence system. The main goal is to help a company make better decisions using data analytics, machine learning, and Power BI dashboards.

The system analyzes demand, delivery delays, inventory levels, supplier performance, transportation efficiency, and supply chain risks.



Supply Chain Analysis

Become a Supply Chain Management Pro !

Dataset

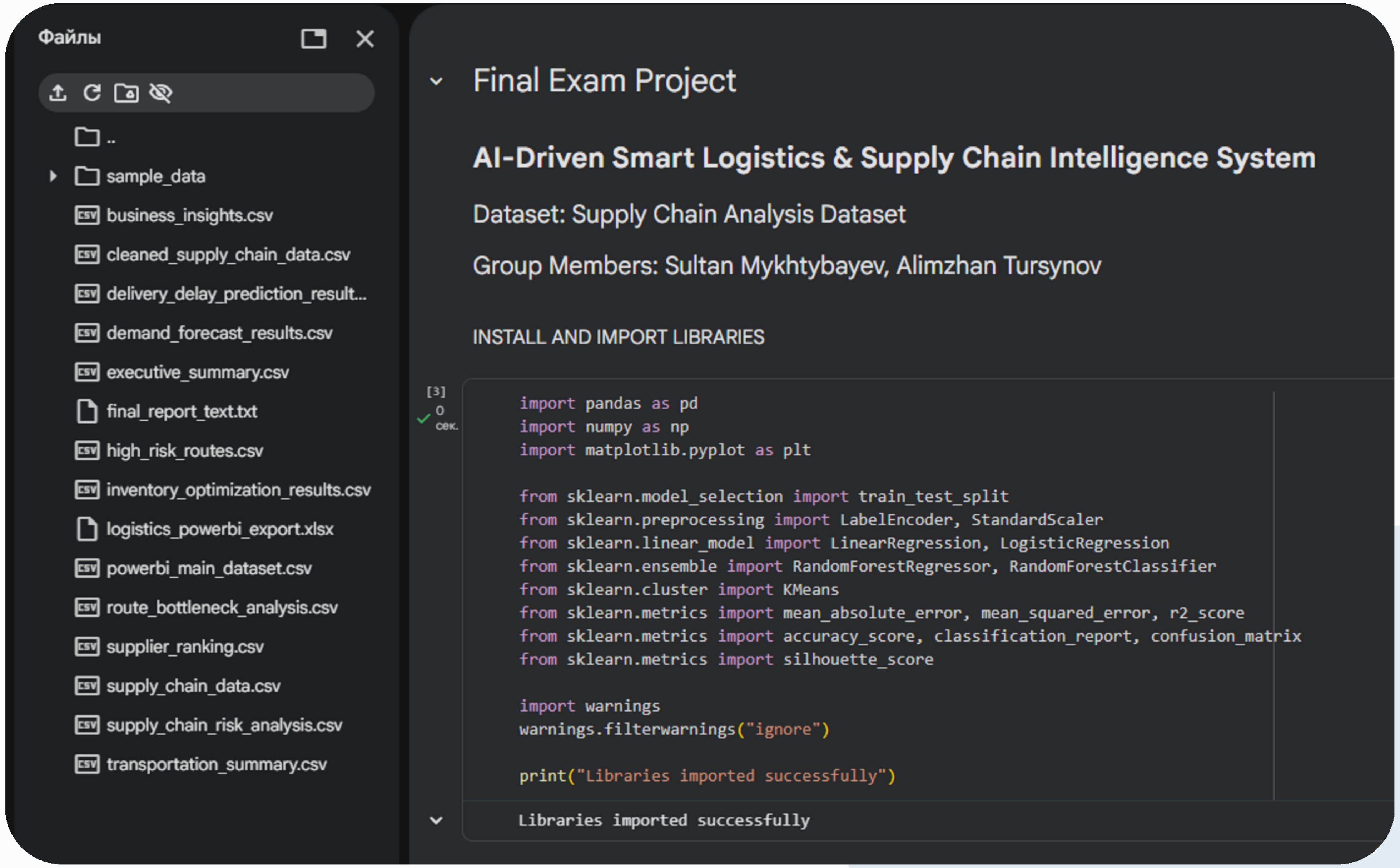
We used the Supply Chain Analysis Dataset from Kaggle.
The dataset includes product type, SKU, price, availability, products sold, revenue, stock levels, lead time, shipping cost, suppliers, transportation modes, routes, manufacturing cost, and defect rates.
In Python, we cleaned the data by removing duplicates, handling missing values, standardizing column names, and creating new calculated columns such as Profit, Delay Status, Inventory Status, Supplier Risk Score, and Transport Efficiency.

Python & AI Analytics

The Python part was completed in Google Colab.

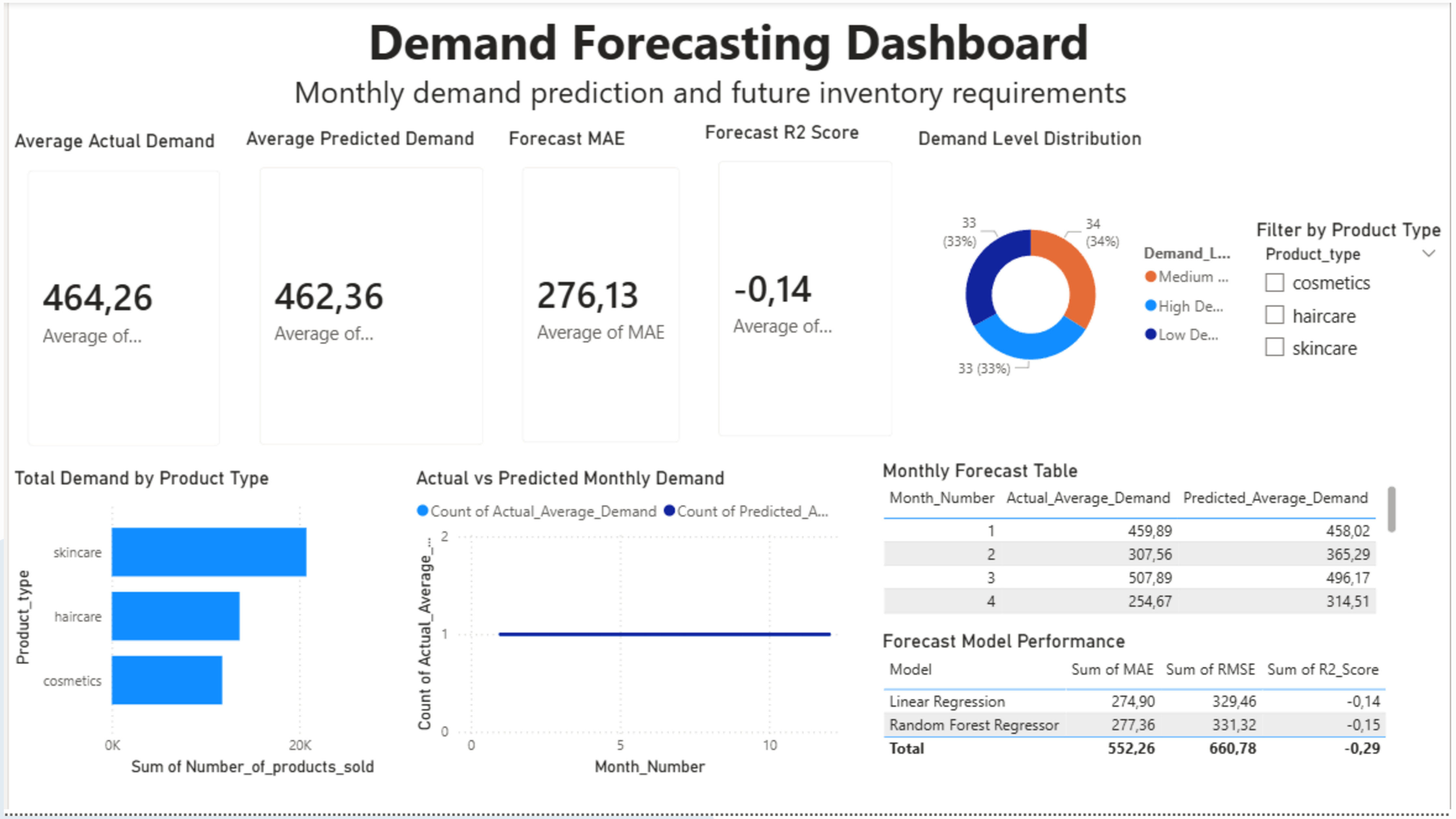
We performed six main tasks:

1. Data cleaning and preparation
2. Exploratory data analysis
3. Demand forecasting
4. Delivery delay prediction
5. Inventory optimization
6. Supplier and transportation analytics



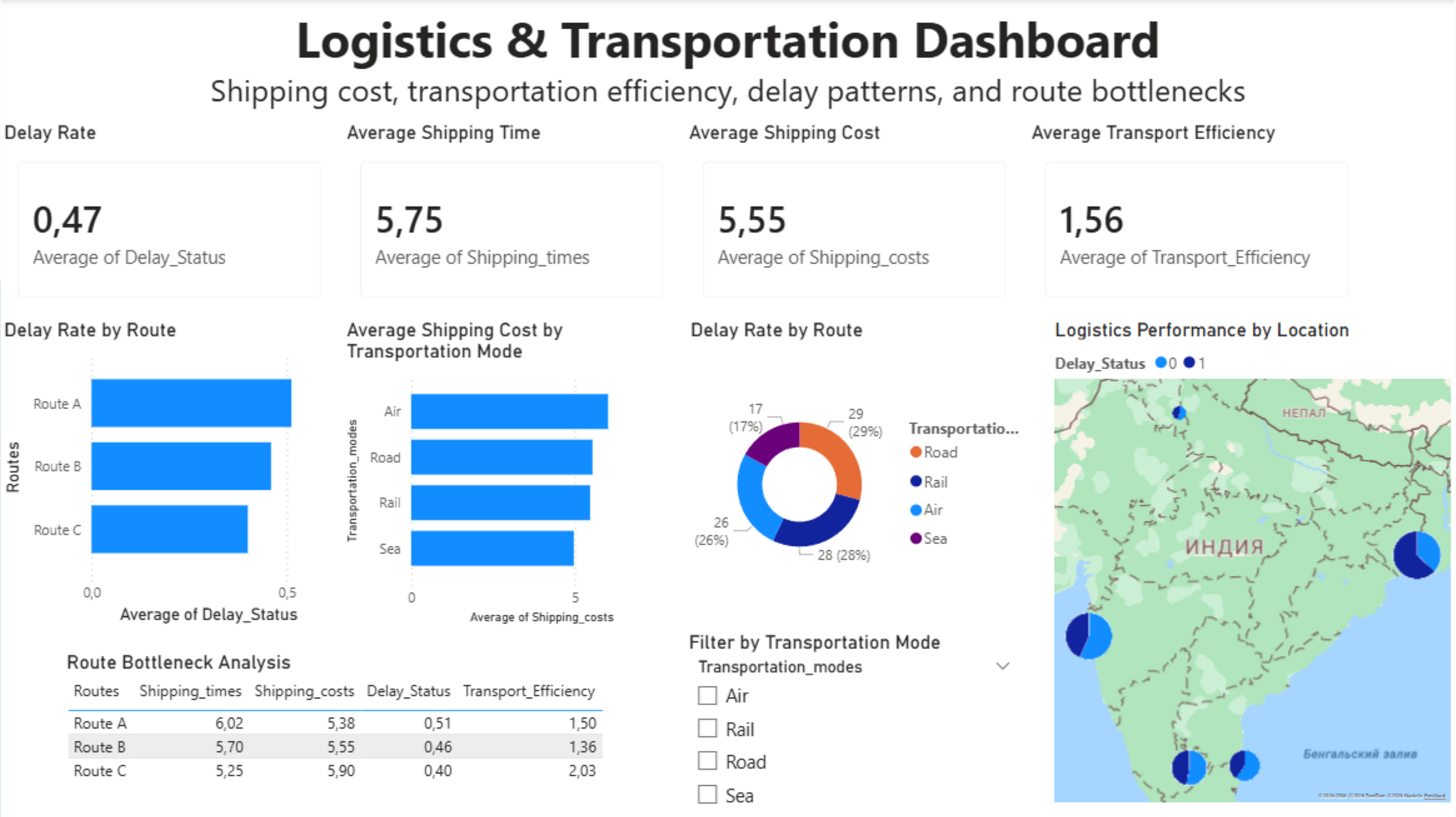
The outputs from Python were exported to Excel and used in Power BI.

Demand Forecasting Model



For demand forecasting, we used machine learning regression models.
The target variable was the number of products sold.
The model predicted future demand and helped estimate future inventory requirements.
We compared Linear Regression and Random Forest Regressor. The forecast results were later visualized in Power BI.
This helps the company prepare enough stock before demand increases.

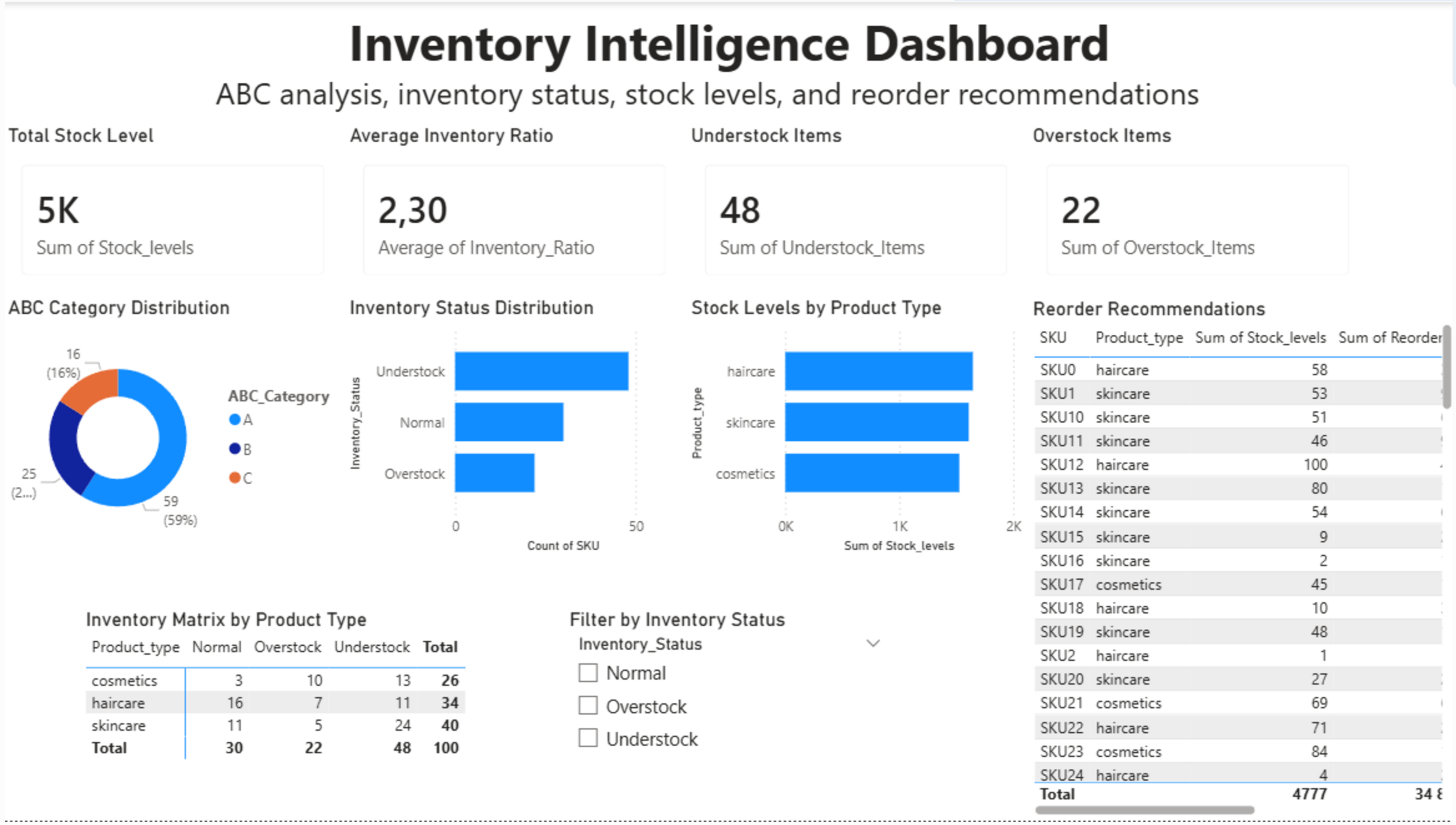
Delivery and Transportation Analysis



This dashboard analyzes shipping cost, delivery time, delay rate, transportation modes, and route bottlenecks.
The delay prediction model helps identify risky deliveries and routes with a higher probability of delay.
The map shows logistics performance by location, while route analysis helps managers find transportation bottlenecks.

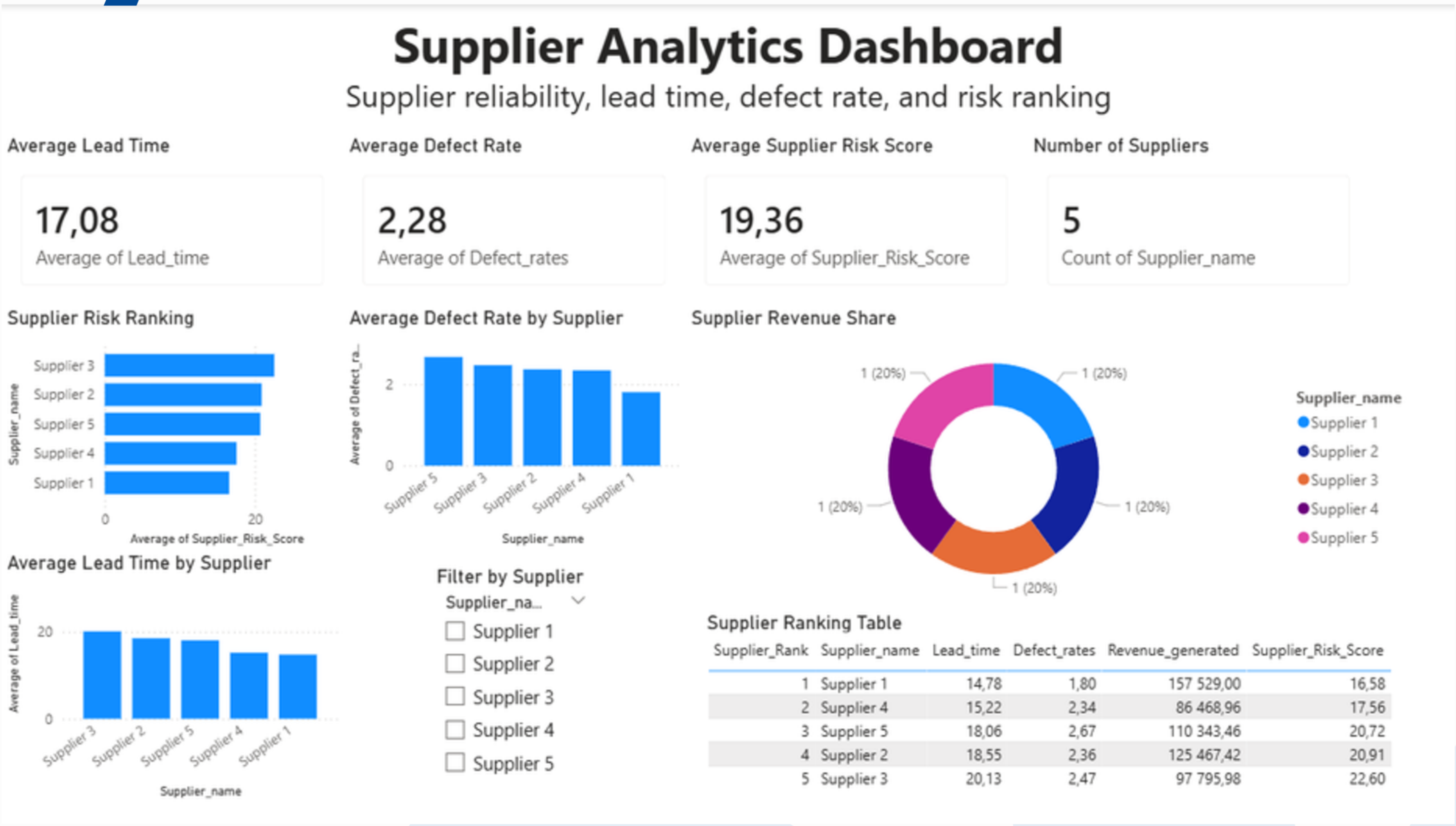
Rating Inventory Optimizationalysis

Inventory optimization was done using ABC Analysis and K-Means Clustering.
ABC Analysis helps identify the most valuable products based on revenue contribution.
The dashboard shows understock, overstock, stock levels by product type, and reorder recommendations.
This helps the company avoid stock shortages and reduce unnecessary inventory costs.
Screenshot / image

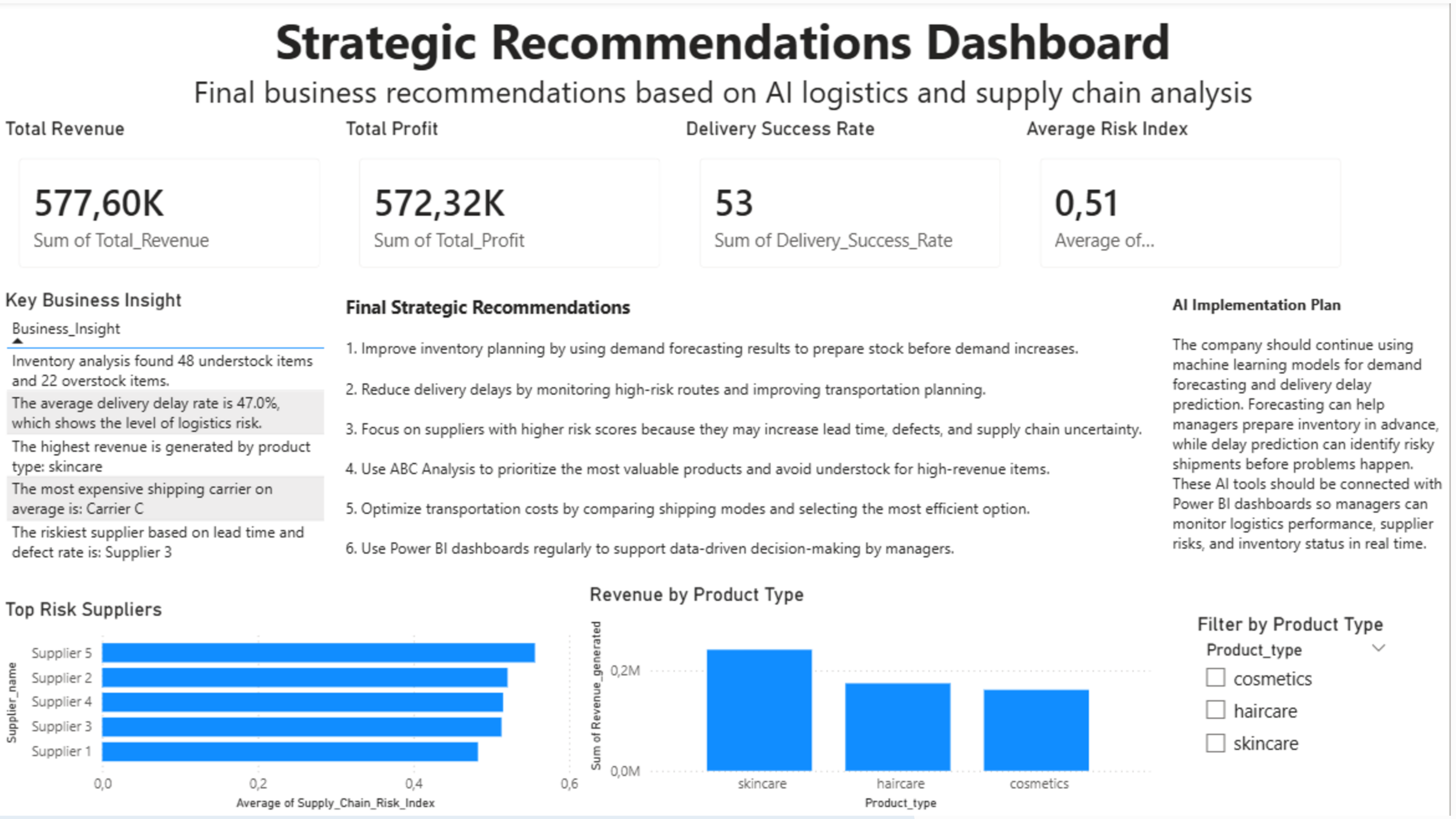


Supplier Analytics and Risk

Supplier analytics was used to evaluate supplier reliability, lead time, defect rate, and supplier risk score.
The dashboard ranks suppliers based on risk.
Suppliers with higher risk scores may cause longer lead times, more defects, and supply chain uncertainty.
Risk analysis also helps identify high-risk routes and suppliers.



Power BI Dashboard and Business Insights



The final Power BI report contains eight dashboard pages.

The main business insights are:

1. Skincare generated the highest revenue.
2. The average delivery delay rate is around 47%.
3. Inventory analysis found 48 understock items and 22 overstock items.
4. Carrier C has the highest average shipping cost.
5. Supplier 3 has the highest supplier risk score.

These insights can help managers improve inventory planning, reduce delays, and optimize supplier and transportation decisions.

Final Recommendations and Conclusion

This project shows how AI and business intelligence can improve logistics and supply chain management.
The company should use demand forecasting to prepare inventory in advance, monitor high-risk routes to reduce delays, focus on risky suppliers, and use ABC Analysis to prioritize important products.
Power BI dashboards allow managers to track logistics performance, supplier risks, inventory status, and strategic recommendations in one place.
Overall, the system supports faster, smarter, and more data-driven supply chain decisions.

Thank You